Pre-Analysis Plan: The Effect of Firstborn Daughters on Gender Attitudes, Political Preferences, and Household Behavior

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1. Aim

This pre-analysis plan outlines a strategy to test whether having a firstborn daughter influences parents' gender attitudes, political preferences, and household behaviors. We leverage the longitudinal structure of the German Socio-Economic Panel (SOEP) in a difference-in-differences framework with individual and time fixed effects.

2. Hypotheses

- **H1:** Having a firstborn daughter leads to more egalitarian gender attitudes.
- **H2:** Having a firstborn daughter leads to more left-leaning political preferences.
- **H3:** Having a firstborn daughter leads to greater gender equality in household behavior.
- **H4:** The effect of having a firstborn daughter on gender attitudes is stronger for men than for women.

3. Data and Sample

Data source: German Socio-Economic Panel (SOEP)

- Individual questionnaires: `PL`, generated variables in `PGEN` (e.g. `pglabgro`, `pgtatzeit`)
- Biography module: `BIOBIRTH` (parent–child links)

Sample restrictions:

- First biological child occurs during panel (from `BIOBIRTH`)
- No births prior to first interview
- At least one year of observations before and after the first birth
- Restrict to biological parents (flag in `BIOBIRTH`)

4. Event Variables

- **Child gender:** `kidsex01` (1 = male, 2 = female)
- **Child birth year:** `kidgeb01` (calendar year)
- **Child birth month:** `kidmon01` (1–12)
- **Daughter_it:** dummy = 1 in year t if `kidsex01` = 2 and $t \geq$ `kidgeb01`

5. Empirical Strategy

We implement an event-study DiD with individual fixed effects α_i and year fixed effects λ_t.

Model for H1–H3

$$Y_{it} \;=\; \alpha_i + \lambda_t \;+\; \sum_{k \neq -1} \delta_k \, \mathbf{1}\{event_time = k\} \;+\; \theta \, \mathbf{1}\{event_time > 0\} \times Daughter_{it} \;+\; \epsilon_{it}.$$

Model for H4

$$\begin{aligned} Y_{it} = \alpha_i + \lambda_t &+ \sum_{k \neq -1} \delta_k \, \mathbf{1}\{event_time = k\} \\ &+ \theta \, \mathbf{1}\{event_time > 0\} \times Daughter_{it} \\ &+ \sum_{k \neq -1} \psi_k \, \mathbf{1}\{event_time = k\} \times Male_i \\ &+ \phi \, \mathbf{1}\{event_time > 0\} \times Male_i \times Daughter_{it} + \epsilon_{it}. \end{aligned}$$

Where:

- Y_{it}: Outcome for individual i in year t.
- α_i: Individual fixed effect.
- λ_t: Year fixed effect.
- $event_time = k$: Years since first birth ($k = -1$ omitted).
- $Daughter_{it}$: 1 if firstborn is a daughter in year t or later.
- $Male_i$: Respondent gender (from `sex` in `PL`).

Standard errors clustered at the individual level.

6. Key Variables and Operationalization

Treatment

- **Child gender**: `kidsex01`
- **Child birth year**: `kidgeb01`
- **Child birth month**: `kidmon01`
- **Daughter_it**: defined above

Outcomes

H1 & H4: Gender-egalitarian attitudes

- `plh0395i01` ... `plh0395i06` (six items on mothers' work & family roles)
- Composite index via mean of z-scores or first principal component

H2: Political preferences

- **Left–right self-placement**: `plh0004` (0 = far left ... 10 = far right)
  - Left-leaning score: $10 -$ `plh0004`
- **Party intention** (robustness): `plh0333` (eligible voters), `ppolpar1`–`ppolpar3` (others)

H3: Household gender equality

- **Time use (weekly):** `pli0043_h` (housework), `pli0046_h` (childcare), `pgtatzeit` (paid work)
- Construct ratio of male/female hours and dummy for "equal" share (each > 40 %)
- **Labor income:** `pglabgro` (gross), `pglabnet` (net, optional)

7. Robustness and Validation

- Test parallel pre-trends ($k < 0$) via event-time coefficients.
- Placebo: pseudo-births assigned to non-parents.
- Add time-varying controls: age, highest education, region (`eastde`), income (`pglabnet`), employment (`pgemplst`).
- Separate models by respondent gender for H4.

8. Estimation and Reporting

- OLS with individual fixed effects.
- Clustered standard errors at individual level.
- Report coefficient tables, event-study graphs, and robustness checks.